

Duggineni Nikhilendra

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EDUCATION

Vellore Institute of Technology

Integrated M.Tech – Computer Science and Engineering (Computational Data Science)

Oct 2022 – Present

CGPA: 8.09 / 10.0

TECHNICAL SKILLS

Core Expertise: Medical Image Processing · Deep Learning · Computer Vision · ML Algorithm Development · Data Science

Languages: Python · SQL · Java · C++ (basics)

Libraries & Frameworks: PyTorch · TensorFlow · Keras · timm · Scikit-learn · NumPy · Pandas · Matplotlib · Seaborn

Computer Vision Tools: pytorch-grad-cam (Eigen-CAM, Grad-CAM) · torchvision · PIL · OpenCV (basics)

Developer Tools: Git · GitHub · Jupyter Notebook · Google Colab · VS Code

Certifications: Applied Machine Learning in Python – University of Michigan | Google Data Analytics Professional Certificate | Oracle Cloud Infrastructure AI

EXPERIENCE

CogniSys AI Private Limited

Data Analytics & ML Intern

Oct 2025 – Dec 2025

Hyderabad, Telangana

- Designed and maintained ETL pipelines to ingest, clean, and transform structured and unstructured datasets, improving downstream data quality for analytics workflows.
- Applied deep learning and statistical modelling techniques within the company's AI development pipeline; collaborated cross-functionally to deliver ML-backed insights.
- Automated recurring data reporting tasks using Python (Pandas), reducing manual effort by over 60% and ensuring report delivery accuracy.
- Recognized for strong analytical thinking and proactive problem-solving across all project deliverables.

PROJECTS

Pneumonia Detection via Swin Transformer & Eigen-CAM | Python, PyTorch, timm, Eigen-CAM, Scikit-learn

- Built a binary chest X-ray classification system using a pretrained Swin Transformer (`swin_base_patch4_window7_224`) trained on the Kaggle Chest X-Ray (Pneumonia) dataset (5,863 images), achieving robust diagnostic performance.
- Implemented advanced data augmentation (random flips, rotation, color jitter, resized crops) and computed class weights to address dataset imbalance, directly improving model fairness and robustness.
- Employed AdamW optimizer with StepLR scheduling and early stopping to prevent overfitting; evaluated model performance using confusion matrix, Precision, Recall, and F1-Score — mirroring real-world medical device validation workflows.
- Integrated Eigen-CAM explainability visualizations on the final attention block of the Swin Transformer, generating interpretable heatmaps that highlight pathological regions in X-ray images — critical for clinical trust and bias analysis.

Financial Data ETL Pipeline & Sentiment Dashboard | Python, PRAW, VADER, TF-IDF, LSTM, SQL

- Built an end-to-end ETL pipeline to extract financial discussion data from Reddit via the Reddit API (PRAW), applying data cleaning and transformation before loading into a structured SQL database.
- Applied VADER sentiment analysis and TF-IDF vectorization to derive investor sentiment scores, enabling quantitative analysis of market mood across thousands of posts.
- Trained an LSTM with Attention model to classify bullish vs. bearish investor sentiment; validated performance using Precision, Recall, and F1-Score.

Automated Misinformation Detection & Reporting System | Python, Scikit-learn, Pandas, TF-IDF

- Developed an automated classification pipeline using a Passive Aggressive Classifier with TF-IDF Vectorization to detect fake news articles with 95%+ accuracy on imbalanced datasets.
- Engineered a complete data preprocessing workflow using Pandas – handling missing values, text normalization, and feature engineering from raw news datasets.
- Benchmarked multiple Scikit-learn classifiers and performed hyperparameter tuning to optimize model performance; reported results via Precision, Recall, and F1-Score metrics.